

Q 1/

Prove mathematically that the following signal is $x(t) = 2t \sin(2000\pi t)$ (8 Marks)

A - Even or odd signal?

sol:

$$x(t) \stackrel{?}{=} x(-t)$$

$$2t \sin(2000\pi t) \stackrel{?}{=} 2(-t) \sin(2000\pi(-t))$$

$$2t \sin(2000\pi t) = (-1)2t(-1)\sin(2000\pi t)$$

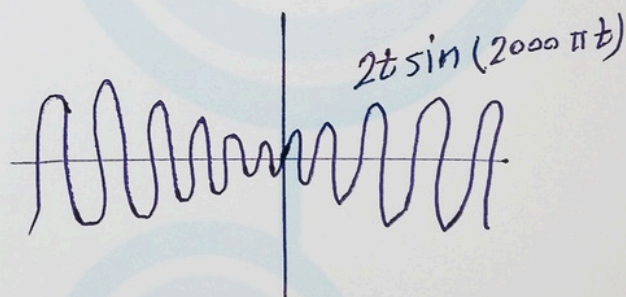
$$2t \sin(2000\pi t) = 2t \sin(2000\pi t)$$

$$x(t) = x(-t) \Rightarrow x(t) \text{ is even } \neq$$

B - Continuous signal or not? and Why?

sol: $x(t)$ is continuous signal because

$$t \in (-\infty, \infty)$$



C - Periodic or aperiodic signal?

$$x(t) \stackrel{?}{=} x(t+T)$$

$$2t \sin(2000\pi t) \stackrel{?}{=} 2(t+T) (\sin(2000\pi(t+T)))$$

$$x(t) \neq x(t+T) \Rightarrow x(t) \text{ is Aperiodic}$$

D- Energy or power signal?

Sol: $x(t)$ is Energy signal because $x(t)$ is Aperiodic signal

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

= number

$$\text{and } P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

= ∞

Q2/ ① $\int_{-5}^3 (3t^3 - 4) \delta(t+1) dt = x(-1)$

$$= 3(-1)^3 - 4 = -7 \neq$$

② $\int_0^3 (3t^3 - 4) \delta(t+1) dt = 0 \neq$

③ $\int_{-3}^3 (3t^3 - 4) \delta'(t+1) dt = (-1)(x'(-1))$

$$= (-1)(9) = -9 \neq$$

④ $\int_{-3}^3 (3t^3 - 4) \delta(2t-2) dt$

$$= \int_{-3}^3 (3t^3 - 4) \left(\frac{1}{2}\right) \delta(t-1) dt$$

$$= \frac{1}{2} \int_{-3}^3 (3t^3 - 4) \delta(t-1) dt$$

$$= \frac{1}{2} x(1) = \frac{1}{2} (3 - 4) = -\frac{1}{2} \neq$$

Q3/ write down the signal in mathematical (6 Marks)

sol:

$$u(t+2) - u(t+1) + r(t+1) - 2(u(t) - r(t-1) + u(t-1) - u(t-2))$$

